



Ali Ibrahim

Full stack software engineer

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■ ABOUT ME

Software engineer with 5 years of experience across backend systems, data engineering, and AI-driven applications. My work spans designing scalable microservices, building data ingestion pipelines, and developing Retrieval-Augmented Generation (RAG) systems using Large Language Models. I have practical experience with technologies such as Qt, React, Spring Boot, and LangChain to build intelligent knowledge retrieval solutions.

■ WORK EXPERIENCE

FULL STACK SOFTWARE ENGINEER – FREIHEIT.COM TECHNOLOGIES GMBH – July 2023 – Current – HAMBURG, GERMANY

Product Data Team

In this team, we were building an e-commerce system for a leading European company in consumer goods and retail. As a member of the Product Data Team, I was responsible for importing data from external source systems, processing it through a pipeline, aggregating it, and delivering it via a REST API to consumers, using a microservices-based architecture. My key responsibilities included:

- Refining, implementing, and maintaining a microservices-based codebase using Kotlin and Spring Boot, migrating an old legacy code written in Java.
- Managing and provisioning infrastructure with Terraform.
- Serving as the team's "Bug Dispatcher," acting as the first point of contact for bug reports, resolving issues, or assigning them to the appropriate developer.

CTS Knowledge Search Team

The project focused on AI-assisted knowledge retrieval for engineering documentation and troubleshooting workflows. We worked with large collections of technical manuals, engineering drawings, support cases, spreadsheets, and internal documentation. My responsibilities included:

- Processing and normalizing data from heterogeneous sources, including documents, spreadsheets, technical drawings, and metadata.
- Designing ingestion pipelines to index and structure data for efficient retrieval.
- Working with Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) approaches to improve knowledge discovery and question-answering over internal datasets.
- Responsible for maintaining the application by contributing to both LangChain backend and React frontend.
- Mentoring and supporting a new team member during his onboarding process.

AnalyticsOne Team

The team focused on harmonizing different workflows to support engineers and researchers in performing analytical calculations on time-series data. My main responsibilities included:

- Assisted a PhD researcher in migrating analytical calculations to the cloud by developing orchestrator and worker services in Python and deploying them using Kubernetes.
- Integrated the developed services into an existing platform built with Java and Spring Boot.
- Contributed to refining the integration of multiple tools within the platform to create more streamlined and unified workflows.

RESEARCH AND DEVELOPMENT ENGINEER – SCASE LTD – November 2020 – December 2022 – LATTAKIA, SYRIA

Sensor Tooling Team

Collaborating on Intenta S-2000 Sensor. Responsibilities:

- Building internal and external tools for sensor integration and configuration using Qt and Angular frameworks.
- Contributed to the development of various services as part of the sensor's firmware.
- Developed robust C++ libraries for integration with sensor tools and services.

■ INTERESTS

Beyond software development, I am passionate about competitive programming and algorithmic problem solving. Over the years, I have participated in numerous local and regional ICPC contests in different roles, including contestant, coach, problem setter, and judge. Some of my most notable achievements include:

- Champion of the local programming contest at my university in 2016, competing among 56 teams.
- Qualified twice as a regional ICPC contestant in 2016 and 2019.
- Coach of a team that qualified for the 2023 ICPC World Finals, held in Luxor, Egypt, in April 2024.
- Served as a judge, problem setter, and problem tester for multiple local and national programming contests in Syria and Egypt.
- Since 2021, led the training and coaching program at my university, including designing learning materials, organizing training camps, and teaching algorithms and problem-solving techniques. These efforts contributed to two teams from my university qualifying for the ICPC World Finals in 2023 and 2024.
- A complete overview of my contributions can be found on my [ICPCID](#).

■ EDUCATION

September 2014 – March 2020 Latakia, Syria

COMMUNICATION AND ELECTRONICS ENGINEERING, Tishreen University

■ SKILLS

React | Spring Boot | LangChain | PostgreSQL | C++ | Python | Kotlin | Java | Go | AWS | GCP | Open Search | GitLab CI/CD | kubernetes | LLM | RAG | Microservices | Problem Solving (Problem Analysis)